

Prop divisor cero no unidad

Proposición 1. Sea R un anillo, entonces $\{\text{divisores de } 0\} \cap \{\text{unidades}\} = \emptyset$.

Demostración: Supongamos por contradicción que $\exists a \in R : a$ es invertible y divisor de 0. Entonces, $\exists b \in R \setminus \{0\} : ab = 0$ y $\exists a^{-1} \in R : aa^{-1} = 1$. Por lo tanto, tenemos que

$$a \cdot b = 0 \implies a^{-1} \cdot a \cdot b = a^{-1} \cdot 0 \implies 1 \cdot b = 0 \implies b = 0. \quad \longrightarrow \longleftarrow \quad \blacksquare$$

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